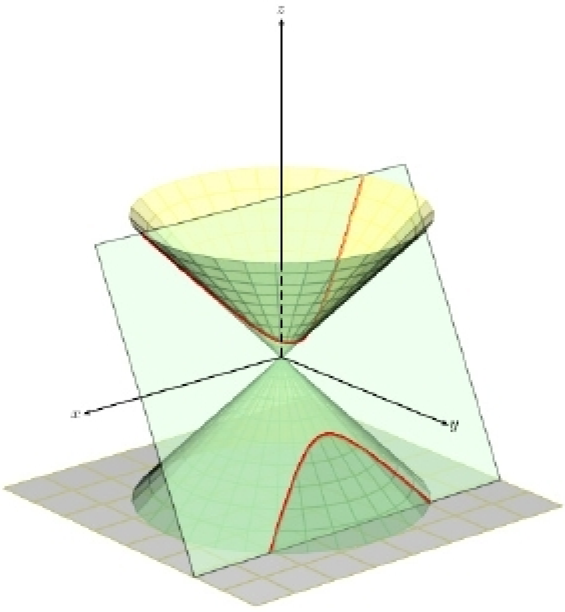
Scientific Graphics Visualisation: An Introduction *to* PSTricks

To conclude the series of three articles on Scientific Graphics Visualisation, let us explore PSTricks, a set of macros that allows the inclusion of PostScript drawings directly in Tex or LaTeX.



In this article, we will discuss PSTricks, a tool used for embedding PostScript graphics directly into LaTeX documents. PSTricks has certain drawbacks like limited output formats and incompatibility with certain LaTeX tools. The immense power offered by PostScript, however, makes it absolutely necessary for us to explore PSTricks. But before that, we need to discuss PostScript a bit.

What is PostScript?

PostScript (PS) is a page description language developed by Adobe Systems. Before the development of PostScript, plotters were used to print vector graphics, and printers were used to print raster graphics. PostScript revolutionised both electronic publishing and desktop publishing by allowing a single machine to handle both vector graphics and raster graphics, thus allowing high quality text and images to be formatted in a single page. Consider the file *post1.ps* with the code shown below:

```
newpath
200 200 moveto
300 200 lineto
300 300 lineto
200 300 lineto
200 200 lineto
2 setlinewidth
stroke
```

The exact meaning of this PostScript code is not important, but learning how this code works will help us understand and appreciate the advantages of PostScript and similar languages. Open the file *post1.ps* with a document viewer capable of displaying PostScript. You will see the image of a square. Later in the article, when we obtain more PostScript files as output of LaTeX scripts, you can open those PostScript files to see human-readable code similar to what we have seen now, but much more complicated. So, this is the philosophy of PostScript — a programming language is used to describe the image and a document viewer interprets that code to display the image. So, even if a PostScript file might appear like a page containing text and images, it actually is a file containing program code.

This approach has led to the huge success of PostScript and similar page description languages. Due to the immense popularity of PostScript, PSTricks was developed as a layer on top of it. Even though the most popular implementation of PostScript is Adobe PostScript, there is also an open source implementation of PostScript called Ghostscript, licensed under the GNU Affero General Public License.

An introduction to PSTricks

PSTricks allows us to include PostScript images directly into LaTeX and TeX documents. Unlike PGF/TikZ, which is a programming language in itself, PSTricks is a set of macros built over the language PostScript. PSTricks was originally developed by Timothy Van Zandt. High quality vector graphics images can be produced by using PSTricks. Vector graphics clearly outperform raster graphics as far as technical diagrams are concerned.

Both the advantage and disadvantage of PSTricks is its dependence on PostScript. PSTricks is very powerful and capable of producing even complex diagrams due to the power of the underlying PostScript language and because of this, PSTricks is more powerful in comparison to PGF/TikZ. But then, this over-dependence on PostScript results in a restricted set of output formats for PSTricks.

In most implementations, the only output format available with PSTricks is PostScript. This again leads to further disadvantages in PSTricks like the inability to work with certain software associated with LaTeX like pdfLaTeX, pdfTeX, etc. So, it is a bit difficult to obtain PDF output files directly from LaTeX files containing PSTricks code. There are ways to work around this and hacks to overcome this problem also. For example, pdfLaTeX can be used to process LaTeX files containing PSTricks with the help of a package called *auto-pst-pdf*. But this is an ugly solution and a better option is to use XeLaTeX instead of pdfLaTeX.